CONSTRUCTION AND VERIFICATION

PA194N5 and PA194N6

Two composed 194-state machines for the consistency of Peano arithmetic

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RESULT

Two new 194-state binary single-tape constructions; no verified reduction to 193 states.

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PA |- (not Halt_blank(PA194Nk) <-> Con(PA)), k = 5, 6.
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The statement uses the inherited proof-producing logical calibration, a uniform no-push command compiler, and new exact source-to-tape checks. No new Lean/Rocq kernel endpoint or iPRA theorem is asserted.

Core verification

194 working states / 388 transition rows per machine

4,816 and 4,877 universal symbolic instruction outcomes

32 simultaneous local fold cases / maximum duration 5

58,320 literal transitions to the complete blank startup frame

The package includes exact native and dense tables, editable TeX, the self-contained JavaScript verifier, inherited proof references, and comparison receipts.

This PDF companion was directly typeset during the execution-runtime outage. The TeX source is supplied separately and has not been compiled in this session.

1. Result and proof scope

We construct two new deterministic single-tape, two-symbol Turing machines, PA194N5 and PA194N6. Each starts on an all-zero two-way infinite tape, has 194 working states, and has 388 transition rows. The designated halt state is excluded from that count. The requested improvement to at most 193 states was not obtained.

The new machines combine two previously separate changes: the no-push successful-modus-ponens controller of PA194N, and the formula-constructor coefficients 5 and 6 of the radix family. The construction changes the command interface and the numerical formula encoding. It preserves the ordinary Cantor encoding of the proof stack and scheduler.

The ordinary mathematical PA argument composes the archived proof-producing logical calibration with a uniform no-push command compiler and new exact source-to-tape certificates. It establishes PA |- [not Halt_blank(PA194Nk) iff Con(PA)] for k=5,6 at the same mathematical proof level as the parents. The inherited PA-to-calculus translation is an explicit dependency. No new end-to-end Lean or Rocq theorem, and no iPRA theorem, is claimed.

All comparisons are pinned to ProveIt commit 1e407ae841e4666f932777fad977ba784ebbcfd5. The repository’s PA README distinguishes these ordinary arguments and finite certificates from unfinished machine-specific formal consistency reductions. This distinction remains in force here.

2. Comparison of the existing designs

The comparison covers 41 placed PA research packets and five current incoming archives: PA194R_research_2026-11-14.zip, pa194N_conpa_2026-11-13.zip, pa194_radix_family_2026-11-13.zip, pa194j_pa194d_conpa_2026-11-13.zip, and pa194q_conpa_2026-11-13.zip.

The older development proceeds from unary-power and 208-state context folds through 196-state source/compiler variants, then PA195, the 195S/195C and M195S8 variants, PA195DC, and two distinct 194-state reader allocations. Larger machines are useful sources of logical and compiler transformations; their state totals cannot be added or subtracted across incompatible interfaces.

Five cached-head PA194C deliveries share one principal dense table, with SHA-256 d759313c03b515293fe06b12769fa1771d3829a2bad89febaa9b4391990f3a4b. Counting packets as independent machines would exaggerate that frontier.

A fresh audit of all .tm, .txt, .tsv and .csv members in the five incoming ZIPs examined 1,331 candidate files, including prose. The exact table recognizer found 171 complete 194-state occurrences, 51 byte representations and 28 action-graph classes. Sixteen classes are report-supported headline or companion designs, nine are supplementary/research tables, and three are explicitly rejected. These are classifications of the supplied evidence, not 28 independent consistency proofs.

The audit allows arbitrary state renaming, every graph-spanning root, left/right reflection, and simultaneous read/write symbol complementation. It compares complete canonical arrays. PA194N5 and PA194N6 match none of the 28 old classes and do not match one another. This audit concerns recognized literal tables; tables existing solely as code or JSON are outside that recognizer.

One cross-packet duplicate is important: the Q packet’s PA194CSZ equals the R packet’s PA194X byte for byte (dense SHA-256 62e93098caf7c09ff2938d07064f0dbbb47c71ec369892968258fff708592269). The report’s frontier table groups them together.

Frontier summary

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Machine Sites Outcomes Startup

PA194-67 3673 4780 58154

PA194-AD 3673 4780 58122

PA194-68 3673 4780 58154

PA194C 3649 4749 57922

PA194CS 3646 4745 57768

PA194N 3654 4755 58154

M4R 3673 4780 58320

M5 3734 4841 58320

M6 3795 4902 58320

PA194J 3646 4755 60062

PA194D 3649 4751 58168

PA194Q 3649 4749 57922

CSZ / X 3646 4746 58014

PA194R 3646 4746 57768

PA194F 3646 4746 57768

PA194A 3646 4746 57978

PA194N5 * 3715 4816 58320

PA194N6 * 3776 4877 58320
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* Newly constructed. Other rows summarize the supplied packets; this table is not an independent formal proof of their calibrations.

3. Exact construction and state accounting

The native parent files are pa194_radix_family/machines/5/machine.native.tm and the corresponding machines/6 file in the pinned radix ZIP. In each parent, replace exactly the row ’164 1 1 R 98’ by ’164 1 1 R 114’. All other native transitions are retained. Native start is 74 and native halt is 254.

Equivalently, rebuild from the primary unfolded 201-control table. Add shared control 330 with rows 0 -> (0,R,61) and 1 -> (1,R,99). Set control 190’s zero row to (0,R,330). Its one row is (1,R,99) for N5 and (1,R,330) for N6. Apply the seven primary reader folds and the additional fold 41 -> 99, whose replacement rows are (0,L,30) and (1,L,2035). Apply the no-push row change above. The supplied verifier checks all eight folds simultaneously against the final table.

The resulting budget is 135 fixed controls plus 67 axiom readers minus 8 compatible folds, giving 194. Every working state occurs in the transition graph reachable from the start label. This is graph reachability, not a claim that every state is visited on the single blank-tape run.

Dense tables number states by breadth-first traversal from native 74, visiting the read-0 successor before read-1. Dense start is 0, working labels are 0 through 193, and halt is 194. Exact tables and native-to-dense label maps are included.

4. Numerical formulas and the uniform logical calibration

Let p(a,b)=(a+b)(a+b+1)/2+b. The stack is [a;w]=p(a,w), with total projections hd and tl. In particular [0;0]=0 and hd(0)=tl(0)=0; numerical stacks are not free lists.

For each fixed k>=4 use I_k(A,B)=k*p(A,B), M_k(a,b)=k*p(a,b)+1, U_k(a,B)=k*p(B,a)+2, and E_k(a,b)=k*p(a,b)+3. Code 0 denotes truth and code 1 is M_k(0,0), the distinguished contradiction. The transposed arguments of U_k are intentional. Negation is N_k(B)=I_k(B,1); conjunction and existential quantification use their usual classical implicational definitions.

The inherited separate-head calculus contains six logical heads, the numerically restricted implicational head rho, every diagonal head Gamma_k(B)=I_k(B,U_k(B,B)), and the guarded separation gate G_k(a,b,c) when c differs from a and b. Its inference rules are modus ponens and theorem generalization. The complete schemas and the archived calibration are retained in the reference TeX fragments.

To transfer radix-four refutations, recursively recode both formula codes and variable labels. Set F_k(0)=0; if n>0 and n=4*p(a,b)+r with 0<=r<4, put F_k(n)=k*p(F_k(a),F_k(b))+r. The decoded components are smaller than n, so course-of-values primitive recursion defines F_k, and PA verifies the recursion. An inverse on the image proves injectivity. This preserves constructor identities, variable binding, the diagonal axiom and the separation guard, and fixes code 1.

Numerical order is not preserved: 40<44 but F_5(40)=75>55=F_5(44). Therefore a restricted rho axiom must not simply be recoded as another restricted axiom. The fixed rho8All derivation in M6/Logic/RestrictedImplication.lean removes that obstruction. Its side conditions use reflexivity and the bounds a,b<=I_k(a,b). Replacing each restricted leaf by this finite unrestricted template gives a primitive-recursive proof translation.

For soundness define a total syntactic translation to arithmetic: implication maps to implication; membership to Bit(x_a,x_b); U to universal quantification; equality to equality; unused residues 4,...,k-1 to closed truth. Every label in the translation of B is smaller than B, so the diagonal head is a vacuity instance. The alias I_k(0,0)=0 is treated using a fixed proof that truth is equivalent to truth implying truth.

PA proves finite binary comprehension for each arithmetic formula theta by induction on a finite bound. The separation gate uses that proof pattern, with a separate coincident-label case a=b. This is a syntactic proof supplier, not a truth predicate or a reflection principle. A translated derivation of code 1 gives a PA refutation. Together with the archived radix-four completeness translation and hereditary recoding, this gives PA-refutation iff C_k-refutation.

5. No-push command semantics and completeness

A command store is (t,w,d,h): visible top, numerical stack, offset, and cached formula. Put y=d+x. The no-push MP_x returns (y,w,y,0) when I_k(t,y)=h, and (0,w,y,0) otherwise. Shift S halts exactly when t+h=1; otherwise it returns (h,tl(w),d,hd(w)).

The axiom command AX_{A,x,y} writes a guarded top g and pushes [q1,...,q9,t;w], then clears d and h. The first eight q entries and g are multiples of k; q9=U_k(d,h) has residue 2 and is at least 2. The diagonal head is separate, so there are nine generated entries plus the saved old top.

These arithmetic facts prove safe selector identities on arbitrary natural stores: AX;S^9 yields (q8,[t;w],0,U_k(d,h)), and AX;S^10 yields (U_k(d,h),w,0,t). Their Shift tests cannot equal 1. The same argument works for k=5 and k=6; it does not assume a nonempty stack.

Write A0=AX_{0,0,0}. Push=A0;S^9;MP_0 sends (t,w,d,h) to (0,[t;w],0,0). Pop=A0;S^11 sends (t,[a;w],d,h) to (t,w,0,a). Discard=MP_0 sends (t,w,0,h) to (0,w,0,0), since its two branches coincide when y=0. Offset_i=MP_i sends (0,w,0,0) to (0,w,i,0), including i=0.

Axiom selectors use Discard;AX;S^(j+1);Pop^(9-j) for q_j, 1<=j<=7. The dominated head uses Discard;Offset_u;AX_{b,a-u,c};S^9;Pop for u<=a. The gate uses Discard;AX;Pop^10. Generalization uses Push;Offset_i;S;AX_{i,0,0};S^10. Each word preserves the arbitrary framed stack and returns with d=h=0.

For modus ponens from a and I_k(a,b), use Discard;Push;C(I_k(a,b));Push;C(a);Push;S;S;MP_b;Pop. The first Push reserves a zero below the two premise computations. Before the two Shifts the store is (0,[a,I_k(a,b),0;w],0,0). The second Shift may halt early if a=1; otherwise MP produces (b,[0;w],b,0), and the final total Pop produces (b,w,0,0). This is valid even when numerical zero pairs collapse.

Induction on a derivation proves that C(p), from any (t,w,0,0), either halts or returns its conclusion with the same stack and d=h=0. For a refutation, append S. Prefix A0 to clear an arbitrary incoming d,h. Thus every refutation supplies a command word that halts from every carried store.

The unchanged scheduler need not reset the store between trials. Prefixing the universally halting word by AX_{j,0,0} produces a cofinal family of fresh candidate codes p(j+2,p(0,p(0,z))), where z is its positive continuation code. The counter starts at two; these candidates are eventually reached unless an earlier trial already halted. Positive continuation pointers strictly decrease. The encoding, proof-word compiler and terminating macro loops have primitive-recursive finite trace witnesses provable in PA.

For the reverse implication annotate t,h and every live canonical stack entry by derivations, or by PA proofs of their arithmetic translations. AX, MP and Shift preserve these annotations. At a halt t+h=1, one of the two visible theorem codes is 1. Its annotation supplies a PA refutation. No annotation is required on the numerical offset.

6. Binding the proof to the literal tables

The generic Cons macro, from registers (T,W,X,Y)=(t,w,0,0), terminates at (I_k(hd(w),t),tl(w),0,0) while preserving other registers. Ordinary stack pairing is unchanged. Only its multiplication loop changes: after i iterations, T=T0+k*i and Y=Y0-i. This proves termination in (k+2)*Y0+1 primitive steps for that loop.

There are 61 Cons occurrences. Their multiplier bodies change from four increments to k increments. The 3,429 sites outside those bodies preserve all operations and successor edges under relocation. After this radix change, the no-push transformation deletes nineteen source instructions: [3695,3714) for N5 and [3756,3775) for N6. Their only exterior entries are respectively 3689 -> 3695 and 3750 -> 3756. Every remaining resolved instruction and successor is checked after relocation.

The retained seven-instruction copy loop sends (T,D,X)=(0,b,0) to (b,b,0), with W unchanged. In the first loop its invariant is (j,b-j,j); in the second it is (b,j,b-j). The exact source cost is 7*b+2. The physical row change enters its existing reader directly.

The supplied JavaScript verifier re-expands the original Lean source DAG, reconstructs both tables, checks all local folds, and symbolically runs every primitive source outcome with arbitrary register values and arbitrary remaining tape frame. Symbolic arithmetic-block summaries are inherited only for 23 controls, whose literal rows are checked unchanged. This is a universal symbolic simulation certificate; its checker step counts are not concrete machine times.

PA194N5 has 3,715 source sites, 4,816 branch outcomes, maximum symbolic duration 499 and total symbolic-checker duration 1,506,840. PA194N6 has 3,776 sites, 4,877 outcomes, maximum 499 and total 1,531,094. All 32 simultaneous fold contexts pass, with maximum local duration 5.

A separate literal blank-tape simulation reaches the complete canonical frame after 58,320 steps in both machines: native state 84, head -2, logical registers (2,0,0,0,0,0,0,0,0), and 26 spare blocks. Equality checks the complete finite set of one-cells, rather than a tape window.

Each source primitive has a positive physical simulation. A source halt therefore yields a physical halt. Conversely, if the physical machine first halted at time T without a corresponding source halt, iterate the totalized source machine for T+1 primitives. Their positive simulations, following the checked startup, would produce a deterministic nonhalting physical trace beyond T, a contradiction. This bounded construction avoids assuming a source program’s infinite execution as an oracle.

Combining the two finite-trace reductions with the command compiler and logical proof translations gives, for k=5,6, PA |- Halt_blank(PA194Nk) iff exists p Prf_PA(p,0=1). Negating both sides gives the requested nonhalting/consistency equivalence.

7. Material difference and further research questions

Graph nonisomorphism alone is supplemented by a command-level distinction. On the legitimate theorem store (t,w,d,h)=(0,3,0,0), where 3=p(2,0) and code 2 is U_k(0,0), MP_0 succeeds. The old radix controller returns (0,9,0,0) because p(0,3)=9; the no-push controller returns (0,3,0,0). The altered stack interface is therefore substantive. The two new machines additionally construct different numerical codes, such as I_5(1,1)=20 and I_6(1,1)=24.

The work reuses the ZFC line’s reserved-zero/no-push idea through the PA194N command compiler. It does not replace PA by the weaker iPRA calculus. ConciseIPRA offers useful proof-checker organization and formalization infrastructure, but equiconsistency with iPRA alone would not establish the requested Con(PA) machine.

Further question 1: can the already-paid triple-increment reader be shared with a suitably reassociated axiom decoder while preserving the eighth fold? A radix-aware search must count helper compatibility, not merely grammar nodes.

Further question 2: can the diagonal-D or premise-exchanged axiom emitters be combined with the radix/no-push controller at an effective reader cost of 58 rather than 59? Initial experiments show that a desired new helper can displace an old compatible fold.

Further question 3: can a certified, size-neutral local transition rewrite expose a ninth useful fold? Such a search should retain an all-context simulation certificate and recheck startup and the complete source interface.

Further question 4: formalize the uniform k-constructor loop, hereditary formula/label recoding, unrestricted rho template, and reserved-zero compiler as reusable Lean lemmas. The current executable certificates do not discharge the missing weak-arithmetic proof endpoints.

8. Digests and reproduction

Run node verify.mjs from the extracted package. Node.js 18 or later is sufficient; the verifier uses no network or third-party package. It regenerates source instructions, rebuilds the transition tables, checks all simultaneous folds, symbolically simulates each source outcome and checks the complete startup tape.

PA194N5 native SHA-256

`e5bbe8edd0c9b204b799fb5b8fa1c7c685ad48276de1e58bb71b8154f5f02216`

PA194N5 dense SHA-256

`b521b1086ad29abdaf213feac2a3d2c0eb1bba7524c36e7f627763aeb2377b09`

PA194N6 native SHA-256

`63b6f604ee34ee119eb92ce31250212572041428 54dec416c6b955f0560b975a`

PA194N6 dense SHA-256

`36da42a81d605ccb8b5edae197c7cde311447bd95b9bf1ab771a3e726a41c6f4`

9. Bounded reduction searches

The primary association search examined 291,067 proposals using the supplied search template, whose fixed controller rows differ from the released primary table. The radix-aware search used the actual released fixed rows and examined 123,117 proposals. Neither supplied a checked improvement below 194.

The local-action search used radius-four tape windows and at most 80 transitions per detour, obtaining 25,707 abstract contexts. All 348 certified neutral one-row alternatives were tested, with a new merge scan after each. No 193-state result was found. Radius six produced 273,587 contexts and no direct additional fold.

The combined quantifier-premise/diagonal-D source was searched for 174,651 rotations, best 197 states. Diagonal-D with radix six was searched for about 78,282 rotations, best 195. Creating the desired extra helper displaced another existing fold.

These are finite sufficient-search languages and heuristic association searches. Their negative outcomes establish no global state-count lower bound.

10. Sources and dependencies

All sources are pinned to ProveIt commit:

`1e407ae841e4666f932777fad977ba784ebbcfd5`

Computability/BusyBeaver/Research/PA/README.md and docs/README.md: formal status, packet placement and 41 earlier deliveries.

PA/docs/pa194_conpa_2026-11-13: primary source grammar, source compiler, seven folds, symbolic executor and separate-head logical calibration.

docs/incoming/pa194N_conpa_2026-11-13.zip: no-push command proof, nineteen-instruction source splice and all-store proof compiler.

docs/incoming/pa194_radix_family_2026-11-13.zip: generic Cons proof and hereditary formula/label recoding, including unused residues and the numerical-order obstruction.

PA/jamesyc-upstream/pa196: James Chang’s source DAG, register engine and M6/Logic/RestrictedImplication.lean.

The reference TeX and Lean files preserve the inherited arguments and their original scope. The root report.tex contains clickable full repository links; the archive includes the applicable licenses.